GlycoTwin • System Architecture

Type 2 Diabetes Physiological Digital Twin & Predictive Decision Support Engine

Happiest Health Challenge 2026

`Multi-Modal Deep Fusion (GRU + MLP)`

1. INGESTION SOURCES

Synthetic Cohort

`500 Pts × 30 d`

- Nathan et al. 2008 calibration (r = 0.98)
- Gamma postprandial curves (30–60m peak)
- Full hypnograms & circadian HR / HRV
- Realistic steps (2,000–12,000/day)

`src/data_gen/`

Wearable Adapters

`Real Devices`

- Abbott FreeStyle Libre (CSV export)
- Dexcom Clarity CGM format
- Apple Health CSV export
- Simulated AI-READI, CGMacros, Shanghai

`src/data_real/`

2. IN-MEMORY VALIDATION

Validator & Sanitizer

`In-Memory`

- Auto-unit conversion (mmol/L → mg/dL)
- 5-minute grid uniform resampling
- Cubic spline imputation (<30m gaps)
- Strict 24h duration gate (≥288 rows)

`src/ingestion/validator.py`

Leakage Auditing

`Zero-Leakage`

- Disjoint patient splits (Train/Val/Test)
- Temporal causality: 24h lookback strictly precedes 2h forecast horizon
- Target: G ≥ 180 mg/dL within 2h given G < 180

`src/features/dataset.py`

3. MULTIMODAL FUSION

Dynamic Sensor GRU

`Lookback`

- Input: [288 timesteps × 5 channels]
- Channels: CGM, HR, HRV, steps, sleep stage
- 2-Layer GRU (64 units, dropout=0.3)
- Final hidden state + LayerNorm → 64-d

`Sensor Branch: 0.9210 AUROC`

Static EHR MLP

`Demographics`

- Input: 27 encoded EHR features
- Demographics, labs, diagnoses, meds, genetics
- 2-Layer MLP (27 → 64 → 32, LayerNorm)
- Cross-modal concatenation head (96-d)

`Fusion Model: 0.9008 AUROC`

4. DIGITAL TWIN ENGINE

PatientTwin State

`Encapsulation`

- Real-time sensor trace & rolling stats
- Calibrated baseline physiological profile
- Active 2h spike probability & risk tier
- Status: Optimal, Watch, or Hyperglycemic

`src/twin/patient_twin.py`

What-If Simulator

`Counterfactual`

- Post-meal brisk walk (GLUT4 translocation)
- Carbohydrate challenge / snack omission
- Sleep hygiene improvement
- Projected risk change (↑/↓ pts)

`PatientTwin.what_if()`

5. CLINICIAN INTERFACE

Decision Dashboard

`Streamlit`

- 10-second Executive Summary & Result Card
- Interactive CGM spline & 2h forecast cone
- Ambulatory Glucose Profile (AGP) modal day
- Clinical Light & Dark Theme (WCAG AA)

`app/streamlit_app.py`

Explainability & Audit

`Occlusion`

- Signed horizontal feature impacts (% risk)
- Scientific Model Report & fair baselines
- F1/Precision/Recall vs threshold curve
- In-memory private drag-and-drop upload

`app/ui_theme.py`